Section D

6. Project

Workout Tracker

Abstract

The Workout Tracker is a simple Microsoft Access based database application designed to record a person's gym workouts and the changes in their body over time. Every training session is stored with its date, start time, duration, the workout performed, the details of the session and the distance covered, so that the user can look back at what was done on any day instead of relying on a notebook.

The database consists of four related tables — Users, BodyHistory, WorkoutLog and WorkoutNameMaint. The Users table holds the personal details of each person, the WorkoutNameMaint table holds the list of workouts such as Abs Day 1, Chest/Tris TT or DDP Energy, and the WorkoutLog and BodyHistory tables record one row for each workout session and each body measurement for a user. The tables are related so that a workout or a measurement always belongs to a valid user and a log entry always refers to a workout that exists in the list.

Records are entered through a set of user friendly forms rather than by typing directly into the tables. The main menu opens the workout log form, the body history form, the user and workout maintenance forms and the reports form, and each form saves its own record. A set of queries — including body history selection, the export queries and the update queries that calculate the energy burnt by a workout — supplies the figures that the reports form needs. The reports form takes a start date, an end date and a workout name and exports the selected workout records for further analysis.

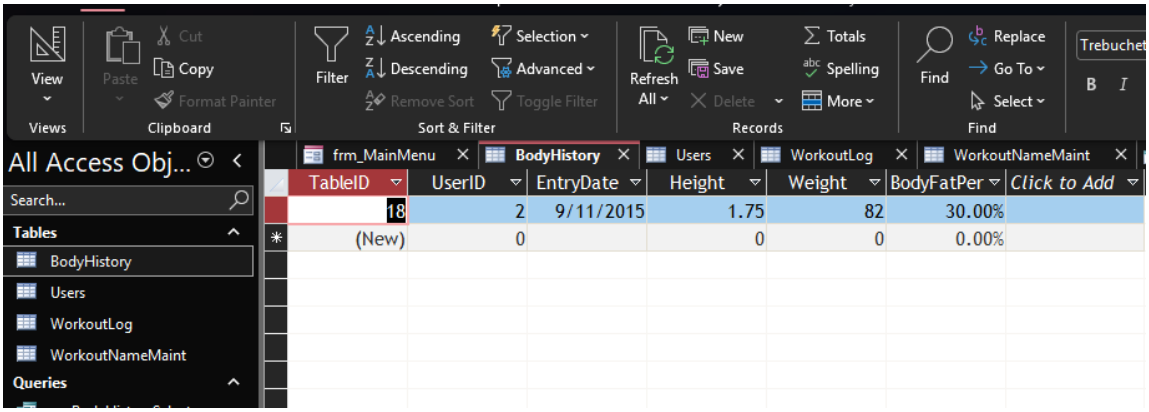
Overall, the project demonstrates the practical use of Microsoft Access for building a small scale information system that is easy to use, keeps the data consistent, and answers common fitness questions without any manual calculation.

Screenshots

Tables

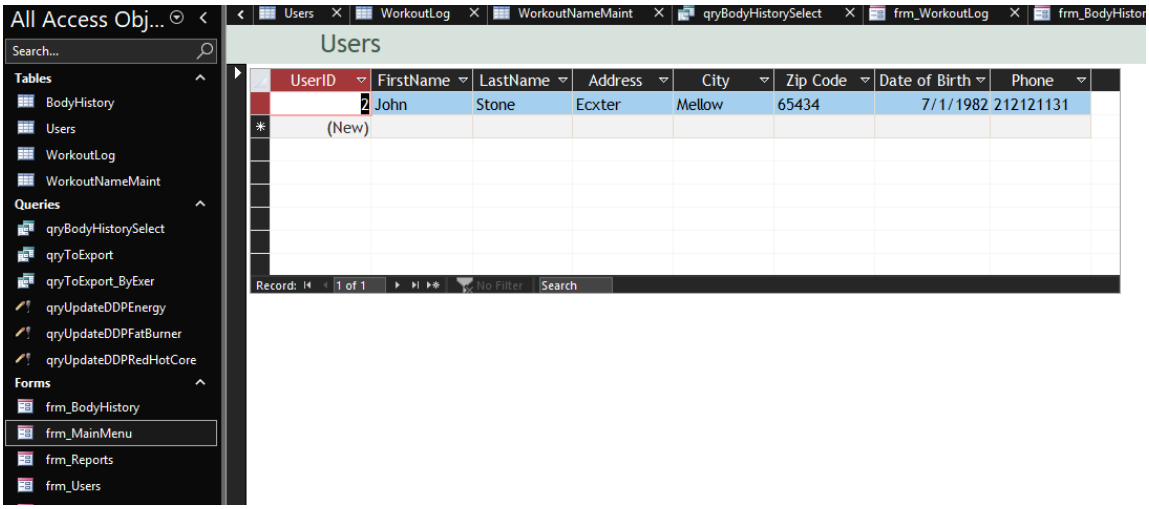
1. Body History Table

The BodyHistory table stores one row for every measurement taken, with the user, the entry date, the height, the weight and the body fat percentage.



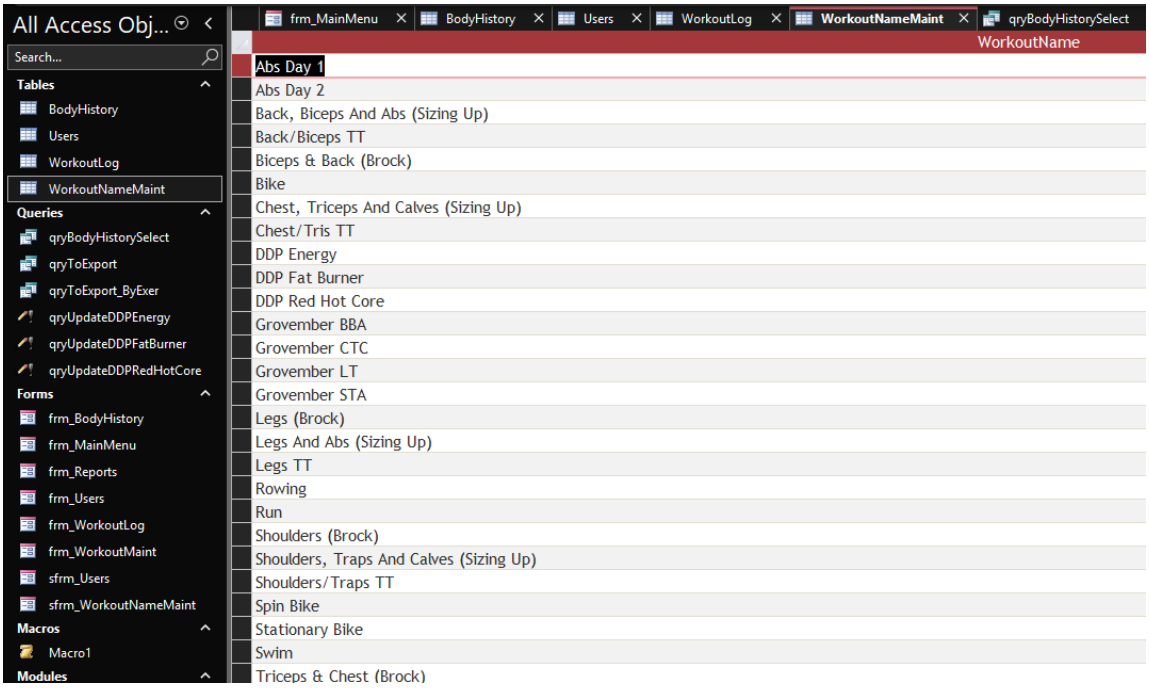
2. Users Table

The Users table stores the personal details of each person using the application — name, address, city, zip code, date of birth and phone number.



3. Workout Name Table

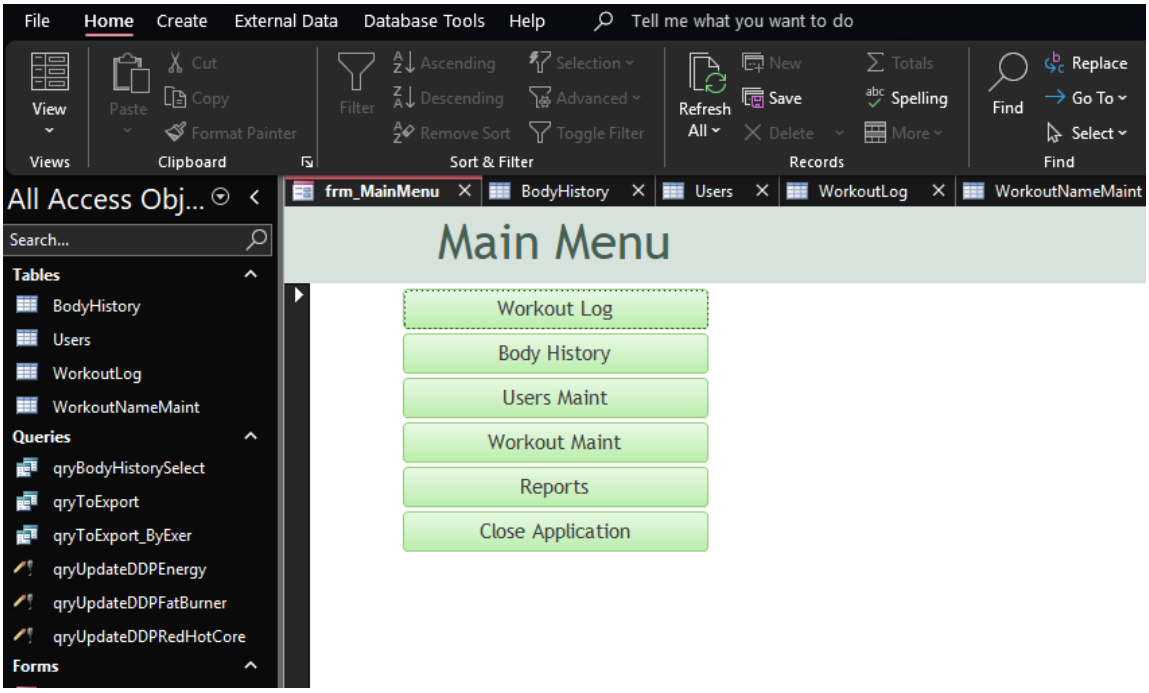
The WorkoutNameMaint table holds the master list of workout names that the log and the reports pick from.



Forms

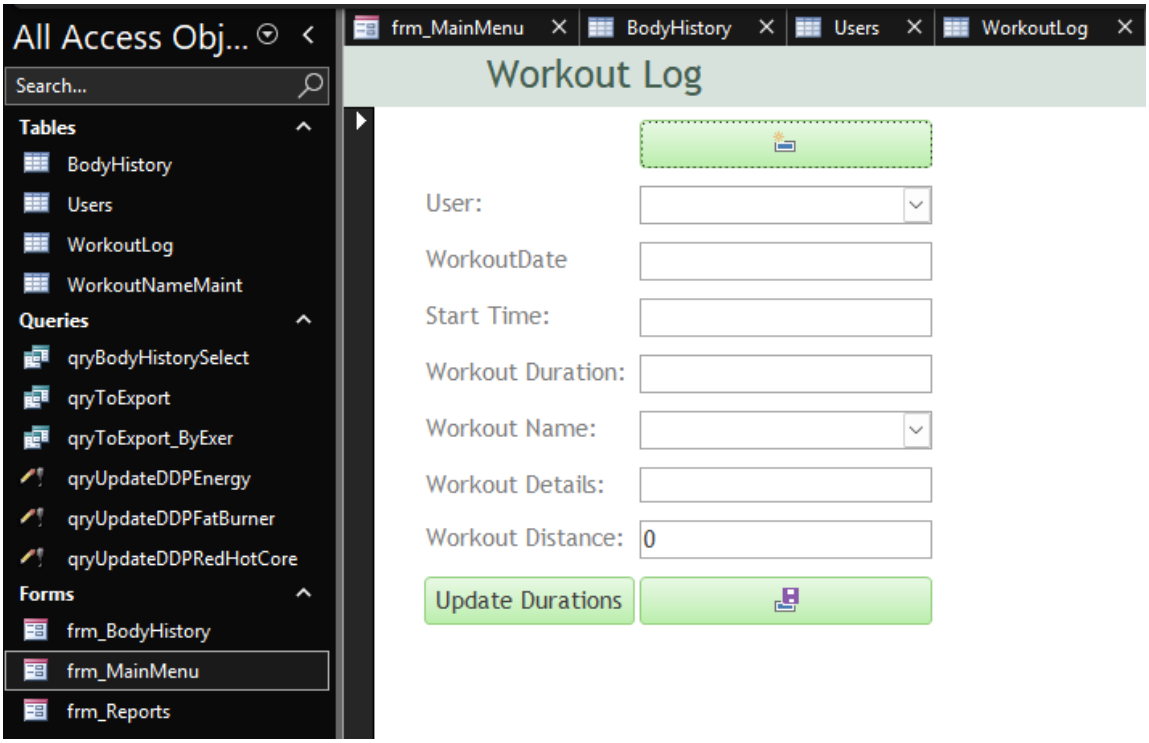
1. Main Menu

The main menu is the starting point of the application. It opens the workout log, the body history, the user and workout maintenance screens and the reports form from a single window.



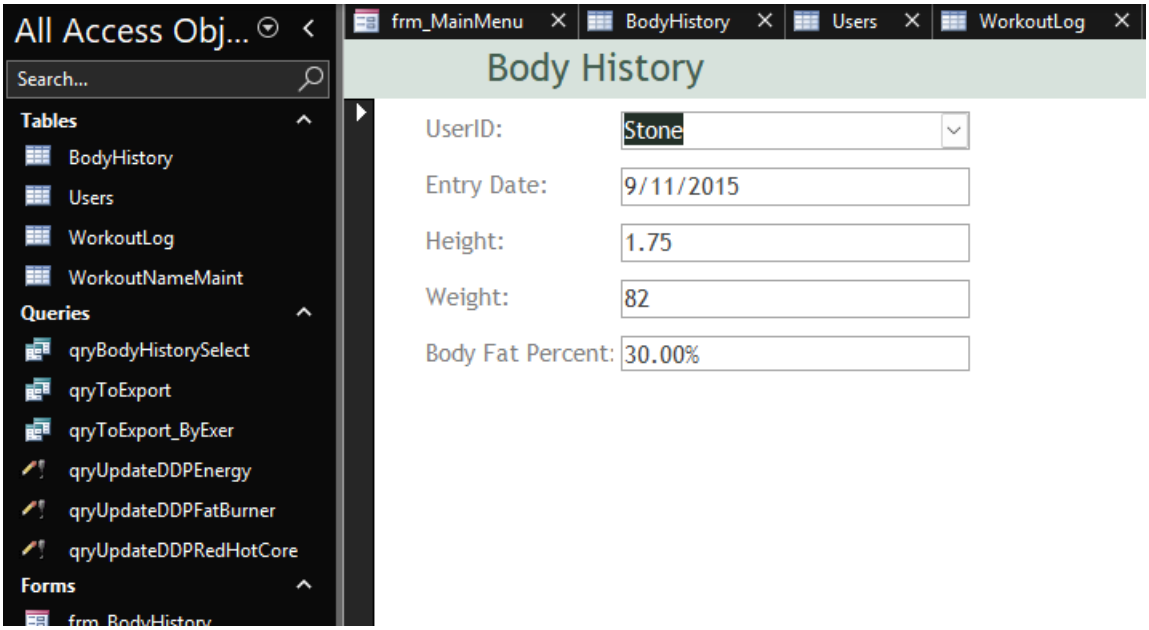
2. Workout Log Form

The workout log form records a training session. The user is picked from a list, and the date, start time, duration, workout name, workout details and distance are stored; the Update Durations button recalculates the duration of the session.



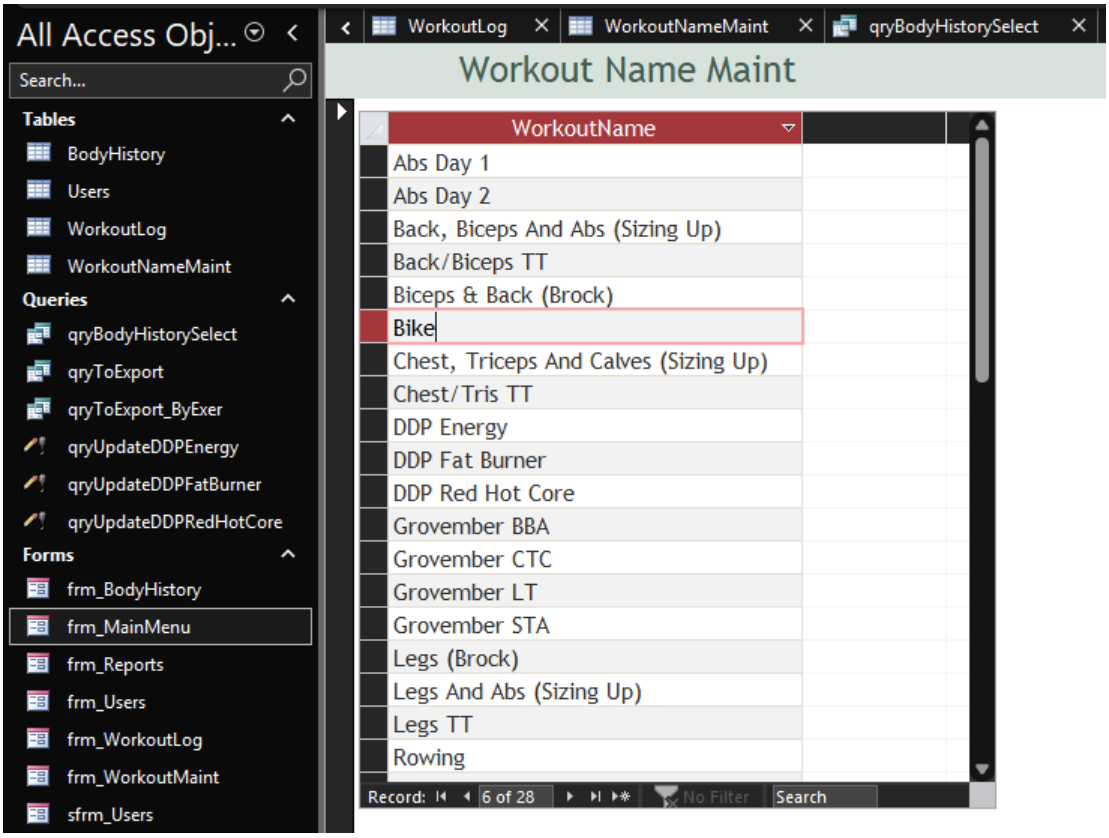
3. Body History Form

The body history form records the height, weight and body fat percentage measured on a given date, so that the progress of the user can be followed over time.



4. Workout Maintenance Form

The workout maintenance form is used to add, edit and delete the workout names used by the log.



Reports

1. Reports Form

The reports form asks for a start date, an end date and a workout name, and exports the matching workout records so that they can be analysed outside the database.

